# Jennifer Olisakwe


## AI/ML Engineer | Backend Engineer | Full-Stack Developer

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## PROFESSIONAL SUMMARY


AI/ML Engineer and Full-Stack Developer with experience building intelligent software systems and scalable web applications using Python, FastAPI, React, PostgreSQL, CI/CD and modern AI technologies. Developed end-to-end healthcare solutions including an AI-powered referral platform that reduced processing time by 45% and a computer vision system that improved radiologist throughput by 2.1×. Skilled in leveraging OCR, speech-to-text, LLMs, and deep learning models to design scalable APIs, secure backend architectures, and cloud-deployed applications that automate workflows and drive measurable operational impact.


## TECHNICAL SKILLS

- Languages: Python, JavaScript, TypeScript, HTML, CSS, Bash, SQL, C++
- Frameworks: FastAPI, Django, React, Next.js, Node.js, SQLAlchemy, Tailwind CSS, Axios, TanStack Query
- AI/ML: OpenAI API, Groq Llama, Whisper, Tesseract OCR, YOLOv8, PyTorch, OpenCV, Grad-CAM, NumPy, Pandas, Jupyter Notebook
- Databases & Cloud: PostgreSQL, MongoDB, Redis, AWS (S3, EC2, Lambda), Render, Vercel, Railway, Supabase
- AI Developer Tools: Cursor, GitHub Copilot, Claude Code, ChatGPT, Windsurf
- DevOps & Tools: Docker, Git, GitHub Actions, CI/CD, REST APIs, Linux, PowerShell, Postman, Jira, Figma
- Testing & Security: Pytest, HTTPX, Jest, Cypress, React Testing Library, JWT, OAuth2, RBAC

## TECHNICAL PROJECTS

### MediFlow - AI Healthcare Referral Management Platform

FastAPI, PostgreSQL, React, TypeScript, Docker, AWS, OpenAI API, Groq Llama 3.1, Whisper, Tesseract OCR, WebSockets

- Architected a healthcare referral platform with 50+ RESTful APIs using FastAPI, PostgreSQL, React, TypeScript, AWS, and Docker, streamlining referral intake and multi-facility clinical workflows.
- Demonstrated expertise in AWS (S3, EC2, Lambda), Tesseract OCR, Whisper, OpenAI, and Groq Llama 3.1 to automate document extraction, transcription, and referral summarization, reducing processing time from 18 minutes to 2 minutes with 92% OCR accuracy.
- Implemented JWT authentication, RBAC, WebSockets, and DICOM-ready storage, contributing to a projected 45% reduction in referral handling time and approximately 360 clinical hours saved annually per clinic.

### OsteoDetect - AI-Powered Bone Fracture Detection System

Python, FastAPI, YOLOv8, OpenCV, Grad-CAM, React, Docker, Vercel

- Engineered an AI-assisted fracture detection platform using YOLOv8, FastAPI, and OpenCV, enabling automated analysis of X-ray images with inference times of approximately 3 seconds per radiograph.
- Implemented Grad-CAM explainability visualizations and confidence scoring to improve clinical interpretability, contributing to an 85% clinician confidence boost during evaluation.
- Built a React-based imaging workflow with REST APIs and automated data prediction pipelines through a streamlined interface, achieving an estimated 2.1× improvement in radiologist throughput.
- Optimized diagnostic workflows through AI-assisted triage, reducing manual review bottlenecks, saving an estimated 40–50 minutes per radiologist per shift, and 78% user satisfaction during testing.

## EDUCATION

**Diploma in Computer Science & Engineering** — 2024

Akirachix — CGPA: 9.06/10